

실습 2 – 202020354 정용훈

OS : MacOS

Terminal

```
// compile
gcc online.c -o online -lpcap
// run
./online
```

Makefile

```
make compile
make run
```

```
(base) jeong-yonghun@Huns-MacBook-Pro exercise2 % make compile
gcc -o online -lpcap online.c
(base) jeong-yonghun@Huns-MacBook-Pro exercise2 % make run
./online
[1] en0 (No description available)
[2] awdl0 (No description available)
[3] llw0 (No description available)
[4] utun0 (No description available)
[5] utun1 (No description available)
[6] utun2 (No description available)
[7] utun3 (No description available)
[8] utun4 (No description available)
[9] utun5 (No description available)
[10] utun6 (No description available)
[11] utun7 (No description available)
[12] lo0 (No description available)
[13] anp1 (No description available)
[14] anp0 (No description available)
[15] en4 (No description available)
[16] en5 (No description available)
[17] en1 (No description available)
[18] en2 (No description available)
[19] en3 (No description available)
[20] bridge0 (No description available)
[21] ap1 (No description available)
[22] gif0 (No description available)
[23] stf0 (No description available)
Enter the interface number (1-23): 1

[Enter the file name(ex : capdump.cap)]: capdump.cap
```

```

982 17:25:09 142.250.207.106 192.168.0.12 UDP 1292 No additional info
983 17:25:09 142.250.207.106 192.168.0.12 UDP 338 No additional info
984 17:25:09 142.250.207.106 192.168.0.12 UDP 996 No additional info
985 17:25:09 142.250.207.106 192.168.0.12 UDP 87 No additional info
986 17:25:09 142.250.207.106 192.168.0.12 UDP 66 No additional info
987 17:25:09 192.168.0.12 142.250.207.106 UDP 120 No additional info
988 17:25:09 192.168.0.12 142.250.207.106 UDP 73 No additional info
989 17:25:09 170.114.194.63 192.168.0.12 UDP 123 No additional info
990 17:25:09 142.250.207.106 192.168.0.12 UDP 1292 No additional info
991 17:25:09 142.250.207.106 192.168.0.12 UDP 343 No additional info
992 17:25:09 192.168.0.12 142.250.207.106 UDP 74 No additional info
993 17:25:09 192.168.0.12 142.250.207.106 UDP 155 No additional info
994 17:25:09 170.114.194.63 192.168.0.12 TCP 66 Src Port: 443, Dst Port: 57651, Seq: 2297140884

```

=== Capture Statistics ===

```

Total packets captured: 1000
IP packet ratio: 99.40%
TCP packet ratio: 45.47%
UDP packet ratio: 54.12%
Bit rate: 13518.40 bits/sec
Packet rate: 24.85 packets/sec

```

[Enter Command (1: next, 0: show all, <num>: specific packet)] : █

[Enter Command (1: next, 0: show all, <num>: specific packet)] : 99

```

=====
99 | 0.013673 | 170.114.194.63 | 192.168.0.12 | 17 | 68 | Info
Internet Protocol Version 4, Src: 170.114.194.63, Dst: 192.168.0.12
    4 .... = Version: 4
    .... 5 = Header Length: 20 bytes
Differentiated Services Field: 0x00 (DSCP: 0x00, ECN: 0x00)
Total Length: 109
Identification: 0x8a35
    2. .... = Flags
...0 .... = Fragment offset: 0
Time to live: 49
Protocol: 17
Header checksum: 0x91e4
[Header checksum status: Unverified]
Source: 170.114.194.63
Destination: 192.168.0.12
[Stream index: 1792721459]
User Datagram Protocol, Src Port: 8801, Dst Port: 51701, Length: 89
Checksum: 0x343b
[Checksum status: Unverified]
Data: 34 bytes
[Enter Command (1: next, 0: show all, <num>: specific packet)] : █

```

```
=====
1090 | 39.472834 | 170.114.194.63 | 192.168.0.12 | 6 | 66 | Info
Internet Protocol Version 4, Src: 170.114.194.63, Dst: 192.168.0.12
    4 .... = Version: 4
    .... 5 = Header Length: 20 bytes
Differentiated Services Field: 0x00 (DSCP: 0x00, ECN: 0x00)
Total Length: 52
Identification: 0xb717
    2. .... = Flags
...0 .... = Fragment offset: 0
Time to live: 49
Protocol: 6
Header checksum: 0x6546
[Header checksum status: Unverified]
Source: 170.114.194.63
Destination: 192.168.0.12
[Stream index: 1792721459]
Transmission Control Protocol, Src Port: 443, Dst Port: 57651, Seq: -1997826412, Ack: 12148555
Header Length: 32 bytes
Flags: 0x0010
Window size value: 4094
Checksum: 0x6def
[Checksum status: Unverified]
Urgent pointer: 0
[SEQ/ACK analysis]
TCP segment data: 32 bytes
(base) jeong-yonghun@Huns-MacBook-Pro exercise2 %
```